

AURA-8 Architecture Diagram & Technical Specification

Economic Times AI Hackathon 2.0 | Problem Statement 8: AI for Industrial Knowledge Intelligence

1. High-Level System Architecture Diagram

The system is structured across 4 decoupled architectural layers (Client Layer, API Orchestration, Intelligence Engine, and External Gemini API):

CLIENT LAYER (Next.js 15 App Router) - Gateway Landing Page (Demo vs Clean Start) - Persistent Floating AI Chatbot Drawer - Interactive 2D HTML5 Canvas Knowledge Graph - Universal Document Ingestor Dropzone	API ORCHESTRATION LAYER - POST /api/upload (PDF stream parsing) - POST /api/query (Gemini RAG call) - GET /api/graph (Knowledge Graph nodes) - POST /api/mode/demo (Sample hydrocracker data)
INTELLIGENCE & VECTOR ENGINE - pdf-parse & PyPDF Text Extractor - Recursive Text Chunker (600ch, 100 overlap) - TF-IDF & Cosine Similarity Vector Index - Regex & NLP Entity Extractor (Tags P-101A, V-301)	EXTERNAL AI LLM SERVICE - Google Gemini API (gemini-flash-latest) - Endpoint: https://generativelanguage.googleapis.com/ - Header: X-goog-api-key - Output: Grounded Q&A; with page citations

2. Ingestion & RAG Execution Sequence Diagram

Step #	Sequence Event	Source → Destination	Description & Payload
1	PDF Document Upload	User → Next.js UI	User drags PDF into dropzone or uploads via file selector.
2	Multipart POST Request	Next.js UI → /api/upload	Posts file FormData to same-origin relative API route.
3	Stream Decompression	/api/upload → pdf-parse	Decompresses /FlateDecode PDF binary stream into clean text.
4	Chunking & Vector Index	/api/upload → Vector Store	Splits text into 600-character overlapping chunks & indexes TF-IDF.
5	Entity & Tag Extraction	/api/upload → Entity Parser	Extracts equipment tags (P-101A) and statutory standards (OISD-137).
6	Graph Population	/api/upload → Knowledge Graph	Populates 2D Knowledge Graph nodes & unlocks dashboard infographics!
7	RAG User Query	User → Persistent Chatbot	User asks technical question in persistent AI chatbot drawer.
8	Cosine Match Search	/api/query → Vector Store	Retrieves top 3 relevant text chunks matching query term frequencies.
9	Gemini LLM Call	/api/query → Gemini API	Posts generateContent prompt with X-goog-api-key header.
10	Grounded Answer Render	Gemini API → Chatbot UI	Renders grounded response with clickable page-level text citations.

3. Technology Stack & Component Responsibilities

Subsystem	Technology Stack	Core Responsibility
Frontend UI	Next.js 15, React 19, Tailwind CSS	Gateway landing page, persistent floating chatbot, interactive Canvas 2D graph
RAG Backend	Next.js API Routes & Python FastAPI	Same-origin /api/upload and /api/query endpoints with zero CORS issues.
LLM Engine	Google Gemini (gemini-flash-latest)	Synthesizing grounded responses, page citations, and executive summaries.
PDF Parser	pdf-parse (Node) & PyPDF (Python)	Decompressing PDF streams, extracting equipment tags (P-101A) and OISD co